



FORTEST

catch the leak.

Protocol documentation for MODBUS-RTU

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Introduction

The purpose of this document is to provide all the technical specifications for communicate with leak testing and flow testing devices of ForTest T-Series and MSeries using the **MODBUS-RTU** layer protocol

The communication type is master / slave. A master unit can be a PLC or a PC for example and the leak test equipment is slave. It is also possible to have a mono master / multi slave topology because every device is identified by an address and every equipment will answer to a specific command request from the master

For understand the meaning of some command it's necessary to have a good acknowledgement of the ForTest device and all the parameters contained in the **TST** , **SET** and **CTP** menu.

Therefore, please refer to the specific user manual for the instrument being used.

All specifications used in this document belong to v.1.1b3 MODBUS Application Protocol

For informations about the **MODBUS** application protocol please refer to the specifications on www.modbus.org

Please note that Modbus RTU over Ethernet is supported only in encapsulated RTU over TCP mode

Software for diagnostic and testing are available at the following [link](#)

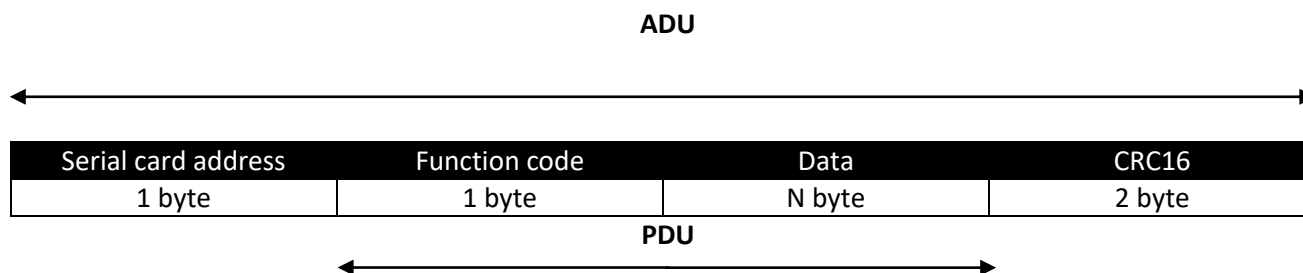
Function code descriptions

The MODBUS-RTU protocol communication by ForTest uses the follow function codes:

- **05 – WRITE SINGLE COILS** - It's used for set/unset some flags on the device for example the start/stop of a test or start/stop the execution of an auto-zero.
- **06 – WRITE SINGLE REGISTER** – It's used to write the value of a parameter into a menu, for example set the value of a filling pressure or the time duration of a test.
- **10 – WRITE MULTIPLE REGISTER** – It's used to write one or more parameter into a menu and to write inside strings. *Available on latest firmware*
- **03 – READ MULTIPLE REGISTERS** – It's used to read the values on the device.

Command Syntax

The **MODBUS** protocol defines a simple protocol data unit (PDU) independent of the underlying communication layers. The mapping of MODBUS protocol on specific buses or network can introduce some additional fields on the application data unit (ADU).



The serial card address field is coded in one byte. Valid codes are in the range of 1 to 255 decimal. The serial card address tell to what device will be sent the PDU information. Is it possible to have a network of MSeries devices and every device must have a serial card address set into his **SET** menu.

It's not possible to have multiple devices with the same serial address on a single serial line.

The function code field of a **MODBUS-RTU** data unit is coded in one byte. Valid codes are in the range of 1 to 255 decimal (the range 128 – 255 is reserved and used for exception responses).

The function code tell what kind of action the receiver will perform. Function code "0" is invalid.

Data code have a custom size in byte depending from the function code to perform.

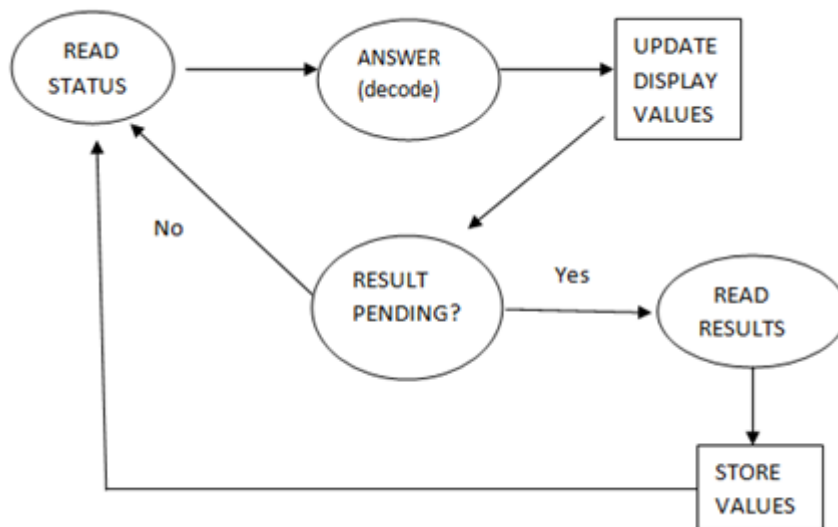
See the paragraph "[Function code description](#)" for more information about data length.

CRC16 is a checksum field for validate the integrity of the packet (PDU + serial card address).

See the paragraph "CRC16" for an example of the CRC16 algorithm used in this MODBUS protocol.

Examples

Example #1 of a full test cycle

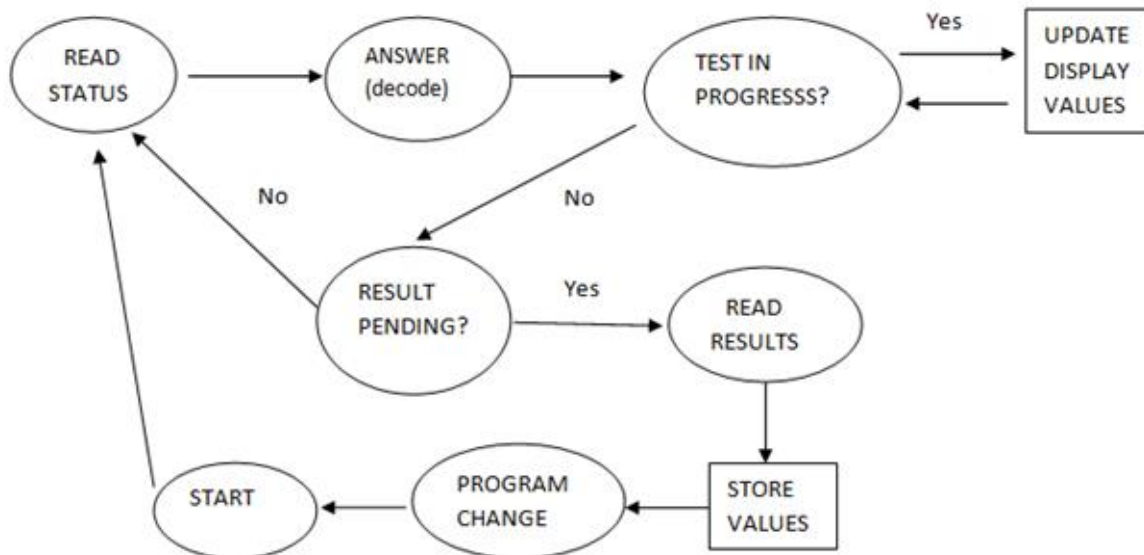


Start of the test from the operator manually

In this example the PLC only use **READ MULTIPLE REGISTERS** command for :

- Read the status of the device in polling
- Read the results if there are pending results

Example #2 of a full test cycle



Start of the test selected from the PLC

In this example the PLC use **READ MULTIPLE REGISTERS** command for :

- Read the status of the device in polling
- Read the results if there are pending results

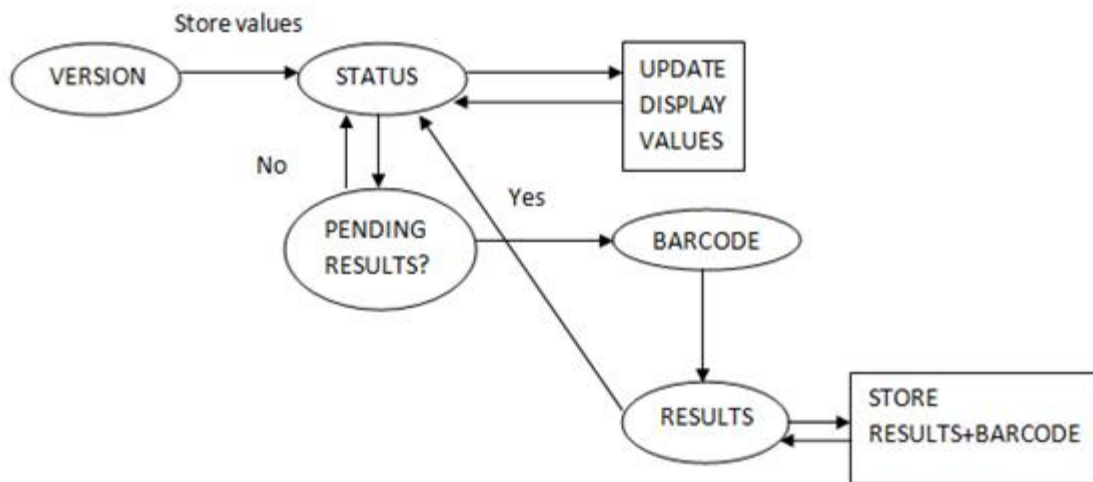
WRITE SINGLE REGISTERS command for :

- Change the current active program

WRITE SINGLE COIL command for :

- Start the execution of the active program

Example #3 of a full test cycle



Start of the test selected from the PLC

In this example the PLC use **READ MULTIPLE REGISTERS** command for :

- Read the version of the firmware
- Read the status of the device in polling
- Read if there is an existing barcode read from the device
- Read the results if there are pending results

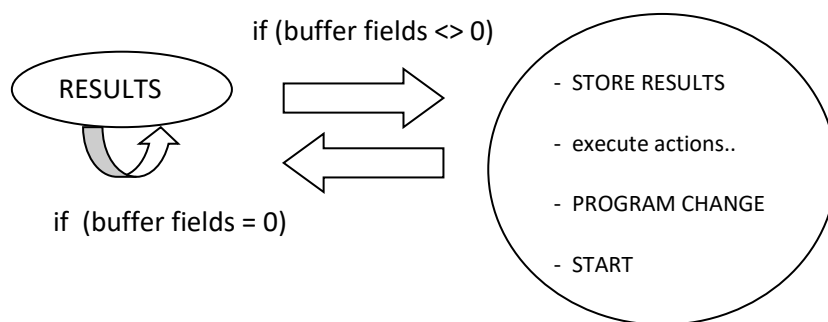
Example #4 The easy way

A simple approach to get test results consist in read the result register in **polling** mode.

When the RX buffer is **empty** (zero byte values) the test is in running and there are not available results.

When the RX buffer is **not empty** the test it's terminated and results can be stored.

New actions can be performed before restarting with the polling sequence.



In this example the PLC use **READ MULTIPLE REGISTERS** command for :

- Read the results if there are pending results

Commands Type

There are 9 types of commands for interact with the instruments.

Four of them are used to require information on the status of the device and these commands are:

- **Status** Command : require the instant state of the device
- **Results** Command : require the last result stored in the device
- **Version** Command : require the firmware version of the device
- **Barcode** Command : require the last barcode read from the device

Five of them are used to perform action on the device and these commands are:

- **Start** a test
- **Stop** a test
- **Auto-zero**
- Reset the count pieces
- Switch on the current active program on the device

To perform the commands it's necessary to compose a correct command with the correct syntax and ask to the device to read or write a defines address of memory. Data will be stored from the PLC and managed.

Single Coil Register Address Mapping

These registers are used to perform actions on the device using the Write Single Coil command (0x05)

For example, is possible to start the active program to perform a test or stop a test in progress

Address hex	Function	CMD Type
0x0A	Start the test (On Bi-channel instruments is referred to the channel A)	Write (0x05)
0x0B*	Start the test on channel B	Write (0x05)
0x0C*	Start the test on channel A and channel B	Write (0x05)
0x14	Abort the test (On Bi-channel instruments is referred to the channel A)	Write (0x05)
0x15*	Abort the test on channel B	Write (0x05)
0x16*	Abort the test on channel A and channel B	Write (0x05)
0x1E	Auto-zero of the device	Write (0x05)
0x40	Reset the CTP	Write (0x05)

* Function only available on Bi-channel instrument

Registers Address Mapping

These registers are used to perform write/read actions on the memory of the Mxxxx device using the Write Holding Register command (0x06) and the Read Holding Register command (0x03)

For example, is possible to read the status of the device, the result of a test or get information on the current firmware loaded.

In case of any issues with the real address mapping, try to use the alias that are the previous ones but with an offset added.

SET parameters of the device

0x3002	Write the value of display contrast regulation
0x3004	Write the value of display minimum lightness
0x3006	Write the value of the serial address
...	...
0x30A3	Serial number (high)
0x30A5	Serial number (low)

TST parameters of the device

0x3322	PROG0: Kind of test
0x3324	PROG0: Filling Pressure
0x3326	PROG0: Tolerance % on the pressure
...	...
0x3360	PROG0: Delay H2 Heater
0x3362	PROG1: Kind of test
0x3364	PROG1: Filling Pressure
0x3366	PROG1: Tolerance % on the pressure
...	...

Please see the tables **TST parameters of the device** , **SET parameters of the device** and

Holding register address mapping for a complete overview of the parameters.

Note: All the requests with wrong values will return an exception code 04.

Holding Registers Address Mapping

These registers are used to perform read actions on the ForTest device. For example is possible to read the status of the device, read the result of a test or get information on the current firmware loaded.

Alias address are used for PLC Siemens when is required a range from 40001 to 49999 and from 400001 to 465535 (Alias feature available only in latest firmware release of T-Series)

Real address hex	Alias address hex	Function	Cmd Type	Reference
0x0030	0x9C41	Read the Status of the device (On Bi-channel instruments is the Status of the channel A)	Read (0x03)	Table 0 - Status Value Mapping
0x0040	0x9C42	Read the Results of the last test executed (On Bi-channel instruments are the Results of the channel A) (This one is erased when a new test start)	Read (0x03)	Table1-ResultsValue
0x0041	0x9C47	Read the Results of the last test executed (This one can be read as the last one on the machine until a new test is completed) (Available only in latest firmware release of T-Series)	Read (0x03)	Table 1.1-ResultValue
0x0042	0x9C48	Read the Results of the last test executed and implements a stack of results; the machine will deletes the last just read result. (Available only in latest firmware release of T-Series)	Read (0x03)	Table 1.1-ResultValue
0x0050	0x9C43	Read the Version of the firmware	Read (0x03)	Table 2 - Version Value Mapping
0x0060	0x9C44	Read the last Barcode read from the device	Read (0x03)	Table 3 - Barcode Value Mapping
0x0060	0x9C44	Write Program Change (Register max value is 299)	Write(0x06)	
0x0070	0x9C45	Read the Status of the channel B (Only available on Bi-channel instruments)	Read (0x03)	Table 0 - Status Value Mapping
0x0080	0x9C46	Read the Results of the latest test executed on the channel B (Only available on Bi-channel instruments)	Read (0x03)	Table1-ResultsValue

SET parameters of the device

These registers are used to perform read/write actions on the general setting of the Mxxx device. For example is it possible to change the contrast regulation and lightness of the display.

Real address hex	Alias Address hex	Function	Size in Bytes	Range values / Reference
0x3002	0x9C57	Display contrast regulation	2 Bytes	0 – 100
0x3004	0x9C58	Display minimum lightness	2 Bytes	0 – 100
0x3006	0x9C59	Serial address of the device	2 Bytes	0 – FF (hex)
0x3008	0x9C5A	Kind of start	2 Bytes	Table – Kind of start
0x300A	0x9C5B	Language	2 Bytes	Table - Language
0x300C	0x9C5C	Graph Visualization	2 Bytes	0 : “OFF” , 1: “ON”
0x300E	0x9C5D	Label signal duration	2 Bytes	0 – 32727
0x3010	0x9C5E	Acoustic signal duration	2 Bytes	0 – 32727
0x3012	0x9C5F	General valve delay closing (only flow test)	2 Bytes	0 – 32767
0x3014	0x9C60	Filter pressure direct	2 Byte	0 – 32767
0x3016	0x9C61	Filter pressure VOUT	2 Byte	0 - 32767
0x3018	0x9C62	Direct pressure dec. fac. vis.	2 Byte	0 – 32767
0x301A	0x9C63	Tolerance % on the opposite decay	2 Bytes	0 – 100
0x301C	0x9C64	Try filling attempts	2 Bytes	0 – 32767
0x301E	0x9C65	Stop test on bad result	2 Bytes	0 : “OFF” , 1: “ON”
0x3020	0x9C66	Equalization during settling	2 Bytes	0 : “OFF” , 1: “ON”
0x3022	0x9C67	COM1 mode	2 Bytes	Table-COM-MODE
0x3024	0x9C68	COM1 baudrate	2 Bytes	Table-COM-BAUDRATE
0x3026	0x9C69	COM2 mode	2 Bytes	Table-COM-MODE
0x3028	0x9C6A	COM2 baudrate	2 Bytes	Table-COM-BAUDRATE
0x302A	0x9C6B	Lock the device with password	2 Bytes	0 : “OFF” , 1: “ON”
0x3036	0x9C71 .	OutAn0 use	2 Bytes	Table-OUT-AN
0x3038	0x9C72	OutAn1 us	2 Bytes	Table-OUT-AN
0x303A	0x9C73	Auto-Zero pressure every	2 Bytes	0 – 32767
0x303C	0x9C74	Recharge at % filling time	2 Bytes	0 – 32767
0x303E	0x9C75	Change scale threshold	2 Bytes	0 – 32767
0x3040	0x9C76	Second branch pressure	2 Bytes	0 – 32767
0x3042	0x9C77	Third branch pressure	2 Bytes	0 – 32767
0x3044	0x9C78	End of cycle signal	2 Bytes	0 – 32767
0x3046	0x9C79	Bad management	2 Bytes	0 – 32767
0x3048	0x9C7A	Abort management	2 Bytes	0 – 32767
0x304A	0x9C7B	Control at % settling time	2 Bytes	0 – 32767
0x307A	0x9C93	Label to print	2 Bytes	0 – 32767
0x307C	0x9C94	Print also linked	2 Bytes	0 : “OFF” , 1: “ON”
0x307E	0x9C95	Print also bad	2 Bytes	0 : “OFF” , 1: “ON”
0x3080	0x9C96	Print also abort	2 Bytes	0 : “OFF” , 1: “ON”
0x3084	0x9C97	Proportional factor	2 Bytes	0 – 32767
0x3086	0x9C99	Derivative factor	2 Bytes	0 – 32767
0x30B4	0x9CB0	Result leds max. brightness	2 Bytes	0 – 32767
0x30B6	0x9CB1	Automatic start	2 Bytes	0 – 32767

0x30C8	0X9CBA	Pressure in economode	2 Bytes	0 – 32767
0x30E2	0x9CC7	DHCP Ethernet (only read) (optional)	2 Bytes	0 : “OFF” , 1: “ON”
0X30E8	0x9CCA	Connection type (only read) (optional)	2 Bytes	<u>Table – Connection type</u>
0x30EA	0X9CCB	DHCP Wireless (only read) (optional)	2 Bytes	0 : “OFF” , 1: “ON”
0x3150	0X9CFE	Serial number high	2 Bytes	0 – 32767
0x3152	0X9CFF	Serial number low	2 Bytes	0 – 32767

TST parameters of the device

These registers are used to perform read/write actions on a specific parameter of a test program. Every parameter cover a fixed address range area. For example the address 0x3322 is the address of the parameter “Kind of test” of the program number 0.

Real Address	Alias Address	Parameter Description	Program Number	Decay	Obstruction	Functional Cycles	Flow Test
0x3322	0x9D00	Kind of test	0	X	X	X	X
0x3324	0x9D01	Filling pressure	0	X	X	X	X
0x3326	0x9D02	Pressure tolerance	0	X	X	X	X
0x3328	0x9D03	Kind of filling	0	X	X	X	X
0x332A	0x9D04	Filling phase duration	0	X	X	-	X
0x332C	0x9D05	Settling phase duration	0	X	X	-	X
0x332E	0x9D06	Test phase duration	0	X	X	-	X
0x3330	0x9D07	Discharge phase duration	0	X	X	-	X
0x3332	0x9D08	Reference: Kind of test	0	Kind of measurement	Min. pres. Obstruction test	-	Min. admitted flow
0x3334	0x9D09	Reference: Kind of test	0	Maximum decay	Max. pres. Obstruction test	-	Max. admitted flow
0x3336	0x9D0A	Reference: Kind of test	0	Piece’s volume	-	-	Pressure>flow compensation
0x3338	0x9D0B	Reference: Kind of test	0	Offset on the decay	-	-	Temperature mass>volume
0x333A	0x9D0C	Bell Time Test **	0	0	0	0	-
0x333C	0x9D0D	Bell Decay **	0	0	0	0	-
0x333E	0x9D0E	Bell Test	0	0	0	0	-

		Cycles **					
0x3340	0x9D0F	Volumetric test time	0	X	O	O	-
0x3342	0x9D10	Tolerance on the measured	0	X	O	O	-
0x3344	0x9D11	Pre-filling pressure	0	X	X	-	-
0x3346	0x9D12	Pre-filling phase duration	0	X	X	-	-
0x3348	0x9D13	Discharge phase duration	0	X	X	-	-
0x334A	0x9D14	Kind of pre-filling	0	X	X	-	-
0x334C	0x9D15	Number of cycles	0	-	-	X	-
0x334E	0x9D16	Work cycle duration	0	-	-	X	-
0x3350	0x9D17	Discharge cycle duration	0	-	-	X	-
0x3352	0x9D18	Program following the running one	0	X	X	X	X
0x3354	0x9D19	Delay on start	0	X	X	X	X
0x3356	0x9D1A	Output coupled with the program	0	X	X	X	X
0x3358	0x9D1B	Input coupled with the program	0	X	X	X	X
0x335A	0x9D1C	Percentage of good	0	X	X	X	X
0x335C	0x9D1D	Measure unit of pressure	0	X	X	X	X
0x335E	0x9D1E	Decay unit measure	0	X	X	x	x
0x3360	0x9D1F	Delay H2 Heater ***	0	O	-	-	-
0x3362	0x9D20	Kind of test	1	X	X	X	X
0x3364	0x9D21	Filling pressure	1	X	X	X	X
0x3366	0x9D22	Pressure tolerance	1	X	X	X	X
0x3368	0x9D23	Kind of filling	1	X	X	X	X
0x336A	0x9D24	Filling phase duration	1	X	X	-	X
0x336C	0x9D25	Settling phase duration	1	X	X	-	X
0x336E	0x9D26	Test phase duration	1	X	X	-	X
0x3370	0x9D27	Discharge phase duration	1	X	X	-	X
0x3372	0x9D28	Reference: Kind of test	1	Kind of measurement	Min. pres. Obstruction test	-	Min. admitted flow

0x3374	0x9D29	Reference: Kind of test	1	Maximum decay	Max. pres. Obstruction test	-	Max. admitted flow
0x3376	0x9D2A	Reference: Kind of test	1	Piece's volume	-	-	Pressure>flow compensation
0x3378	0x9D2B	Reference: Kind of test	1	Offset on the decay	-	-	Temperature mass>volume
0x337A	0x9D2C	Bell Time Test **	1	O	O	O	-
0x337C	0x9D2D	Bell Decay **	1	O	O	O	-
0x337E	0x9D2E	Bell Test Cycles **	1	O	O	O	-
0x3380	0x9D2F	Volumetric test time	1	X	O	O	-
0x3382	0x9D30	Tolerance on the measured	1	X	O	O	-
0x3384	0x9D31	Pre-filling pressure	1	X	X	-	-
0x3386	0x9D32	Pre-filling phase duration	1	X	X	-	-
0x3388	0x9D33	Discharge phase duration	1	X	X	-	-
0x338A	0x9D34	Kind of pre-filling	1	X	X	-	-
0x338C	0x9D35	Number of cycles	1	-	-	X	-
0x338E	0x9D36	Work cycle duration	1	-	-	X	-
0x3390	0x9D37	Discharge cycle duration	1	-	-	X	-
0x3392	0x9D38	Program following the running one	1	X	X	X	X
0x3394	0x9D39	Delay on start	1	X	X	X	X
0x3396	0x9D3A	Output coupled with the program	1	X	X	X	X
0x3398	0x9D3B	Input coupled with the program	1	X	X	X	X
0x339A	0x9D3C	Percentage of good	1	X	X	X	X
0x339C	0x9D3D	Measure unit of pressure	1	X	X	X	X
0x339E	0x9D3E	Decay unit measure	1	X	X	x	x
0x33A0	0x9D3F	Delay H2 Heater ***	1	O	-	-	-
0x33A2	0x9D40	Kind of test	2	...	..	..	..
...	...	...	...	...	...	...	...
0x33E2	0x9D60	Kind of test	3				
...	...	...					
0x3422	0x9D80	...	4				
...	...	...					
0x3462	0x9DA0	...	5				
...	...	...					

0x3F62	0xA320	Kind of test	49	X	X	X	X
...		...	...				
0x3FA0	0xA33F	Delay H2 Heater	49	O	-	-	-
0x413D	0xA340	Kind of test	50	..	..	..	..
...		...	...	...	...	..	..

X Possible

- Not possible

** If the device is **compliance (M8980...)**

*** If the optional **hydrogen** is enabled

For a complete list of all offsets between programs please refer to the table **TST PROGRAM MAPPING**

List of readable and writable strings

Available only in latest
firmware release of T-Series

The following parameters can be written and read with multiple read and write commands

Description	Size	Address	Alias address
Program name (0) #	16 characters	0xEA74	0xC281
Program name (1) #	16 characters	0xEA75	0xC282
Program name (2) #	16 characters	0xEA76	0xC283
...	...	...	...
Program name (100) #	16 characters	0xEAD8	0xC2E5
...	...	...	**
Program name (299) #	16 characters	0xEB9F	**
Company name #	16 characters	0xEBAA	0xC2E6
Operator name #	16 characters	0xEBAB	0xC2E7
Lot code #	16 characters	0xEBAC	0xC2E8
Ethernet Ip Address *	4 characters	0xEBAD	0xC2E9
Ethernet Gateway *	4 characters	0xEBAE	0xC2EA
Ethernet Subnet *	4 characters	0xEBAF	0xC2EB
Wi-Fi Ip Address *	4 characters	0xEBB0	0xC2EC
Wi-Fi Gateway *	4 characters	0xEBB1	0xC2ED
Subnet Wi-Fi *	4 characters	0xEBB2	0xC2EE
Wi-Fi Access point name*#	16 characters	0xEBB3	0xC2EF
Wi-Fi password *#	16 characters	0xEBB4	0xC2F0
Hotspot name **	16 characters	0xEBB5	0xC2F1
Hotspot password **	16 characters	0xEBB6	0xC2F2

* Only readable address, to write these strings see [Writing Network and Ethernet Strings with the command \(0x10\)](#)

Strings that are codified in ascii code, example: 0x30 = 0 , 0x41 = A etc..

** Strings that don't have alias address

Commands

01 (0x05) Write Single Coil

This function code is used to write a single flag as ON. A value of 0xFF00 request set the output to ON state. All others values are illegal and will not affect the output.

Request

Description	Size in Bytes	Value
Function Code	1 Byte	0x05
Output Address	2 Bytes	0x0000 to 0xFFFF
Output Value	2 Bytes	0x0000 or 0xFF00

Response

Description	Size in Bytes	Value
Function Code	1 Byte	0x05
Output Address	2 Byte	0x000A to 0x0028
Output value	2 Byte	0x0000 or 0xFF00

Output Address Mapping

Address hex	Address decimal	Function
0x0A	10	Start the test
0x14	20	Abort the test
0x1E	30	Autozero of the device
0x28	40	Reset of CTP

Note: with the Write Single Coil command only the 0xFF00 values will be accepted. All the requests with wrong values will return an exception code 04.

Example

TX	01 05 00 0A FF 00 AC 38	→	Set the address 0x0A to ON, the device will start a test
RX	01 05 00 0A FF 00 AC 38	→	ACK of the request
TX	01 05 00 14 FF 00 CC 3E	→	Set the address 0x14 to ON, the device will abort a test
RX	01 05 00 14 FF 00 CC 3E	→	ACK of the request

02 (0x06) Write Single Register

This function code is used to write a single holding register in a remote device. The request PDU specifies the address of the register to be written. The normal response is an echo of the request confirming that contents have been written.

Request

Description	Size in Bytes	Value
Function Code	1 Byte	0x06
Register Address	2 Bytes	0x0000 to 0xFFFF
Register Value	2 Bytes	0x0000 or 0xFFFF

Response

Description	Size in Bytes	Value
Function Code	1 Byte	0x06
Register Address	2 Bytes	0x0000 to 0xFFFF
Register Value	2 Bytes	0x0000 or 0xFFFF

Example

TX 01 **06** 30 04 00 64 C6 E0

→ Set the display light regulation to the value 100

RX 01 **06** 30 04 00 64 C6 E0

→ ACK of the parameter set

TX 01 **06** 30 02 00 05 8A C9

→ Set the display contrast regulation to the value 5

RX 01 **06** 30 02 00 05 8A C9

→ ACK of the parameter set

TX 01 **06** 00 60 00 02 08 15

→ Change the current program number to 2

RX 01 **06** 00 60 00 02 08 15

→ ACK of the program change

NOTE : All TST and SET parameters are stored into a EEPROM memory so a defined number of written are allowed by this technology. Please limit the number of write and use the "program change" command to switch between different test parameters.

Available only in latest
firmware release of T-Series

03 (0x10) Write Multiple Register

This function code is used to write a block of contiguous registers (1 to 123 registers) in a remote device. The requested written values are specified in the request data field. Data is packed as two bytes per register. The normal response returns the function code, starting address, and quantity of registers written.

Request

Description	Size in Bytes	Value
Function Code	1 Byte	0x10
Starting Address	2 Bytes	0x0000 to 0xFFFF
Quantity of Registers	2 Bytes	1 to 32 (0x20)
Byte Count	1 Bytes	2 x N*
Register Value	N*2 Bytes	0x0000 or 0xFFFF

N = Quantity Of Registers

Response

Description	Size in Bytes	Value
Function Code	1 Byte	0x10
Starting Address	2 Bytes	0x0000 to 0xFFFF
Quantity of Registers	2 Bytes	1 to 32 (0x20)

Example

TX 01 10 33 22 00 02 04 00 00 13 88 3D C9 → Set the kind of test to decay and the filling
pressure to 500.0

RX 01 10 33 22 00 02 EE 86 → ACK of the parameter set

TX 01 10 33 26 00 02 04 00 14 00 00 71 68 → Set the pressure tolerance to 20 % and the kind
of filling to Time

RX 01 10 33 26 00 02 AF 47 → ACK of the parameter set

TX 01 10 33 2A 00 02 04 00 64 00 C8 71 70 → Set the filling phase duration to 10.0 Second and
the settling phase duration to 20.0 Second

RX 01 10 33 2A 00 02 6F 44 → ACK of the parameter set

Writing Strings with the command (0x10)

Available only in latest
firmware release of T-Series

For writing strings with the Modbus RTU protocol, we created a list of sequential alias addresses and every one of these represent a string.

Request Syntax

- The starting address is the alias of the chosen string (for the table with all the string address see Readable and Writable Strings)
- The quantity of register is the number of character to write inside the string divided by 2 (The maximum width of each string is written on the table Readable and Writable Strings))
- The Byte count is the quantity of register multiplicated for 2
- The content of the string must be written in the request data field and converted in ascii code .
(if the number of the character is odd the last byte must be 0)

Response Syntax

The syntax of the response is the same as the standard command (0x10).

Example

TX 01 10 EA 74 00 03 06 54 65 73 74 20 31 10 33 → Set the Program name of the Program 1 to
"Test 1"

RX 01 10 EA 74 00 03 F4 0A → ACK of the parameter set

TX 01 10 EB AA 00 04 08 46 6F 72 54 65 73 74 00 A1 5C → Set the Company name to
"ForTest"

RX 01 10 EB AA 00 04 D4 0E → ACK of the parameter set

Writing Network and Ethernet Strings with the command (0x10)

Network and Ethernet configuration strings (Ex: IP address, Gateway, Subnet etc....) can't be written individually and works in a different way.

Request Syntax

Available only in latest
firmware release of T-Series

- The starting address must be 0xEBAD (C2E9 Alias)
- The quantity of register must be 23
- The byte count must be 46
- Request data field:
 - 1 byte for the Connection type that we want to configure: (0 Wi-Fi, 1 hotspot, 2 Wi-Fi-ForTest, 3 Ethernet)
 - 16 bytes in ascii code for the Wi-Fi or the Access point name, in case of Ethernet those bytes will be ignored (if the length of the name is less than 16 characters, the remaining bytes must be at zero)
 - 16 bytes in ascii code for the password of the Wi-fi or the Access point, in case of Ethernet those bytes will be ignored (if the length of the password is less than 16 characters, the remaining bytes must be at zero)
 - One byte for the mode of the DHCP (0 Off 1 On)
 - 4 Bytes for the Ip address, in case of active DHCP those bytes will be ignored
 - 4 bytes for the gateway, in case of active DHCP those bytes will be ignored
 - 4 bytes for the subnet, in case of active DHCP those bytes will be ignored

Response Syntax

The syntax of the response is the same as the standard command (0x10)

Example

TX 01 10 EB AD 00 17 2E 00 4E 65 74 20 31 00 00 00 00 00 00 00 00 00 00 30 31 32 33 34 35
36 37 38 39 00 00 00 00 00 00 00 00 00 C0 A8 00 03 C0 A8 00 01 FF FF FF 00 07 AF

→ Set:

- Connection type to Wi-Fi
- Wi-fi name: Net 1
- Wi-fi password : 0123456789
- DHCP OFF
- IP: 192.168.0.3
- Gateway: 192.168.0.1
- Subnet: 255.255.255.0

RX 01 10 EB AD 00 17 24 02

→ ACK of the parameter set

04 (0x03) Read Holding Registers

This function code is used to read the contents of a contiguous block of holding register in a remote device. The request PDU specifies the starting register address and the number of registers to be read.

Request

Description	Size in Bytes	Value
Function Code	1 Byte	0x03
Starting Address	2 Bytes	0x0000 to 0xFFFF
Quantity of Registers	2 Bytes	1 to 125 (0x7D)

Response

Description	Size in Bytes	Value
Function Code	1 Byte	0x03
Byte Count	1 Byte	2 x N
Register Value	N x 2 Bytes	

N = Quantity Of Registers

Example

ID	FC	SA_H	SA_L	QR_H	QR_L	CRC_H	CRC_L
01	03	00	30	00	20	44	1D

TX 01 **03** 00 30 00 20 44 1D → Read status (0x30) of the device

RX 01 **03** 40 00 20 00 00 00 00 00 0D 00 00 00 00 00 01 00 02 00 24 00 00 00 00 00 00 00 00 3C 00
02 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 02 00 00 00 26 00 53 00 C1 00 FF 00
00 00 00 CD FD

ID	FC	SA_H	SA_L	QR_H	QR_L	CRC_H	CRC_L
01	03	00	30	00	01	84	05

TX 01 **03** 30 02 00 05 2B 09 → Read first 5 parameters of SET menu

RX 01 **03** 0A 00 20 00 20 00 01 00 02 00 00 00 B5

Reading Strings with the command (0x03)

Available only in latest
firmware release of T-Series

To read a string in Modbus RTU the command to use is the (0x03), strings can be read one at time using their address as the starting one and the number of character divided by two as the quantity of register (if the number of character is odd, add 1 to make it even).

Request Syntax

- The starting address is the alias of the chosen string (The addresses are written on the table Readable and Writable Strings)
- The quantity of register is the number of character to read inside the string divided by 2 (It's advisable to read the maximum size of each string that is written on the table Readable and Writable Strings, to not occur in any data loss)

Response Syntax

The syntax of the response is the same as the standard command (0x03)

Example

TX 01 **03** EB AA 00 08 51 C8 → Read the company name in the setup menu

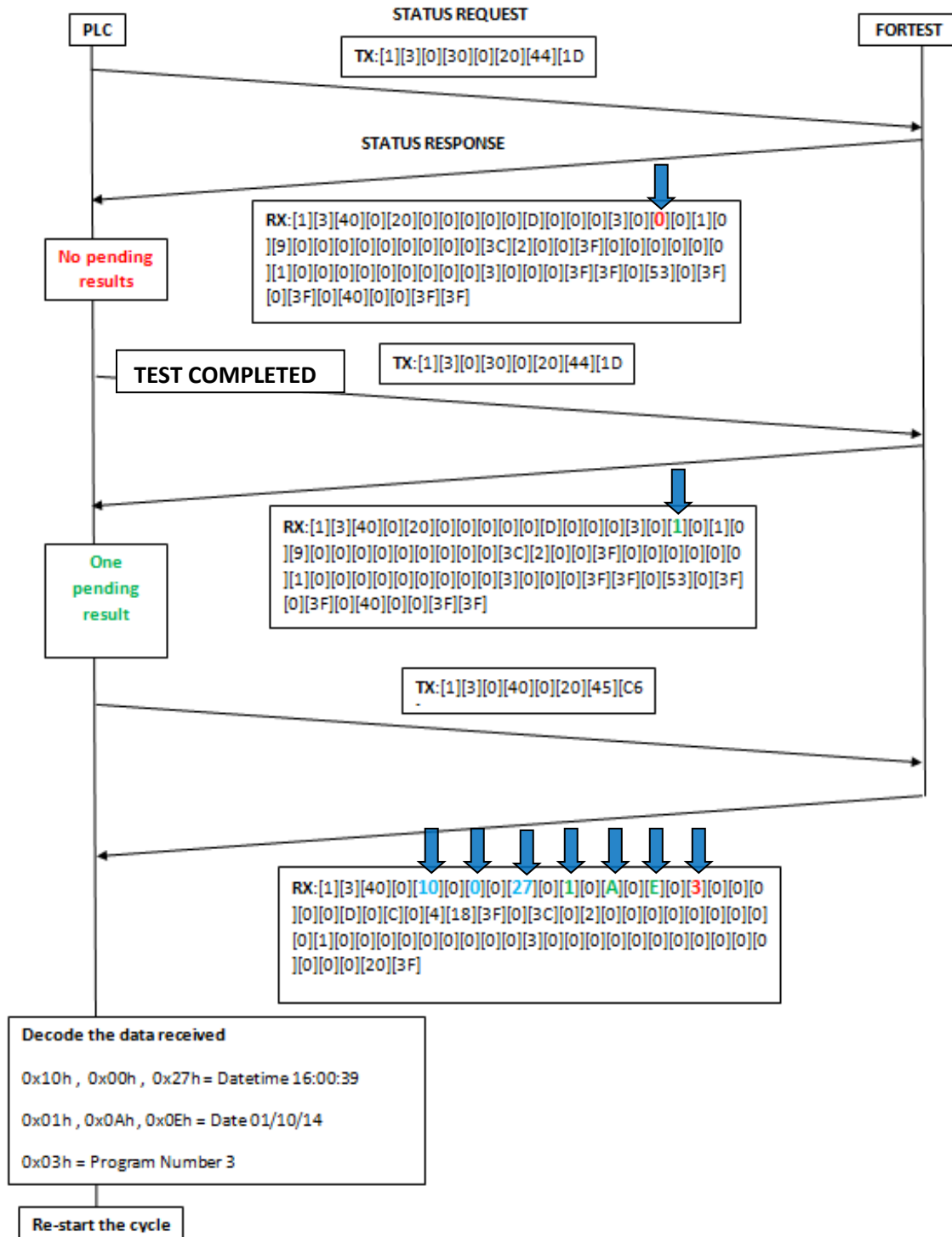
RX 01 **03** 10 43 49 41 4F 00 00 00 00 00 00 00 00 00 00 B7 E0 → ACK of the parameter set

Tx 01 **03** EA 74 00 08 30 0E → Read the program name of the program

Rx 01 **03** 10 50 52 47 30 00 00 00 00 00 00 00 00 00 00 0C EF → ACK of the parameter set

Example of data exchange

The following example uses both instantaneous **status** read commands and read **results** commands



See the [Results Table Mapping](#) to get a complete overview of data obtained with this command

Results request in polling mode

[illegible]

Tables Reference

Table - Status Value Mapping

Register Position (dec)	Definition	Size Bytes	Range	Reference
1	Errors mask	2 Byte	0-65535	-
3	Last active status on the device *	2 Byte	0-65535	Table-Status
5	Last active sub-status on the device *	2 Byte	0-65535	Table-Sub status
7	Last active result phase on the device *	2 Byte	0-65535	Table-Results
9	AUX state – not used *	2 Byte	0-65535	-
11	Program Number active on the device	2 Byte	0-65535	-
13	Number of pending results on the device	2 Byte	0-65535	-
15	Index menu of the last parameter modified	2 Byte	0-65535	-
17	Index of the last parameter modified	2 Byte	0-65535	-
19	Index sub-menu of the last parameter modified	2 Byte	0-65535	-
21	Index last parameter modified on the sub-menu	2 Byte	0-65535	-
23	Time remaining end of test (HIGH) *	2 Byte	0-65535	-
25	Time remaining end of test (LOW) *	2 Byte	0-65535	-
27	Unit measure of time	2 Byte	0-65535	-
29	Decimal points of time *	2 Byte	0-65535	-
31	Sign of the pressure *	2 Byte	0-65535	-
33	Pressure at end of test (HIGH) *	2 Byte	0-65535	-
35	Pressure at end of test (LOW) *	2 Byte	0-65535	-
37	Unit measure of pressure at end of test	2 Byte	0-65535	-
39	Decimal points pressure at end of test	2 Byte	0-65535	-
41	Sign of decay *	2 Byte	0-65535	-
43	Decay at end of test (HIGH) *	2 Byte	0-65535	-
45	Decay at end of test (LOW) *	2 Byte	0-65535	-
47	Unit measure of the decay at the end of test	2 Byte	0-65535	-
49	Decimal points decay at end of test	2 Byte	0-65535	-
51	Sign temperature C°	2 Byte	0-65535	-
53	Temperature Value (HIGH)	2 Byte	0-65535	-
55	Temperature Value (LOW)	2 Byte	0-65535	-
57	Unit measure of the temperature	2 Byte	0-65535	-
59	Mask 8 IN	2 Byte	0-65535	-
61	Mask 8 OUT	2 Byte	0-65535	-
63	Mask 4 IN, 4 OUT of the expansion	2 Byte	0-65535	-

* Parameters that change between channel A and channel B (This feature is only available on Bi-channel instruments)

Table - Results Value Mapping

Register Position (dec)	Definition	Size Bytes	Range	Reference
1	Hour execution of the test *	2 Bytes	0-65535	-
3	Minute execution of the test *	2 Bytes	0-65535	-
5	Seconds execution of the test *	2 Bytes	0-65535	-
7	Day execution of the test *	2 Bytes	0-65535	-
9	Month execution of the test *	2 Bytes	0-65535	-
11	Year execution of the test *	2 Bytes	0-65535	-
13	Program Number *	2 Bytes	0-65535	-
15	Chained test	2 Bytes	0-65535	0 : OFF , 1 : ON
17	Kind of Test *	2 Bytes	0-65535	Table-Kind of test
19	Result of the test *	2 Bytes	0-65535	Table-Results
21	Phase of the test *	2 Bytes	0-65535	Table-Phase of test
23	Time remaining end of test (HIGH) *	2 Bytes	0-65535	-
25	Time remaining end of test (LOW) *	2 Bytes	0-65535	-
27	Unit measure of time *	2 Bytes	0-65535	Table-Measure
29	Decimal points of time *	2 Bytes	0-65535	-
31	Sign of the pressure *	2 Bytes	0-65535	0 : "+", 1 : "-"
33	Pressure at end of test (HIGH) *	2 Bytes	0-65535	-
35	Pressure at end of test (LOW) *	2 Bytes	0-65535	-
37	Unit measure of pressure at end of test *	2 Bytes	0-65535	Table-Measure
39	Decimal points pressure at end of test *	2 Bytes	0-65535	-
41	Sign of decay *	2 Bytes	0-65535	0 : "+", 1 : "-"
43	Decay at end of test (HIGH) *	2 Bytes	0-65535	-
45	Decay at end of test (LOW) *	2 Bytes	0-65535	-
47	Unit measure of the decay at the end of test *	2 Bytes	0-65535	Table-Measure
49	Decimal points decay at end of test *	2 Bytes	0-65535	-
51	Sign of AUX1 decay	2 Bytes	0-65535	0 : "+", 1 : "-"
53	AUX1 Decay at end of test (HIGH)	2 Bytes	0-65535	-
55	AUX1 Decay at end of test (LOW)	2 Bytes	0-65535	-
57	Unit measure of the AUX1 decay end of test	2 Bytes	0-65535	Table-Measure
59	Decimal points AUX1 decay at end of test	2 Bytes	0-65535	-
61	Sign of AUX2 decay	2 Bytes	0-65535	0 : "+", 1 : "-"
63	AUX2 Decay at end of test (HIGH)	2 Bytes	0-65535	-
65	AUX2 Decay at end of test (LOW)	2 Bytes	0-65535	-
67	Unit measure of the AUX2 decay end of test	2 Bytes	0-65535	Table-Measure
69	Decimal points AUX2 decay at end of test	2 Bytes	0-65535	-

* Parameters that change between channel A and channel B (**This feature is only available on Bi-channel instruments**)

The result command can read a **maximum** of 35 registers and a **minimum** of 1 register

TX 01 03 00 40 00 08 45 D8 → Read the last available result

RX 01 03 10 00 0A 00 18 00 2C 00 16 00 07 00 0E 00 00 00 00 22 77

Table –Results Value Mapping (only for 0x41 and 0x42 holding register)

Available only in latest
firmware release of T-Series

Byte Index (dec)	Definition	Size Bytes	Range	Reference
1	Lost Results	2 Bytes	0-65535	-
3	Results not yet transmitted	2 Bytes	0-65535	-
5	Hour execution of the test *	2 Bytes	0-65535	-
7	Minute execution of the test *	2 Bytes	0-65535	-
9	Seconds execution of the test *	2 Bytes	0-65535	-
11	Day execution of the test *	2 Bytes	0-65535	-
13	Month execution of the test *	2 Bytes	0-65535	-
15	Year execution of the test *	2 Bytes	0-65535	-
17	Program Number *	2 Bytes	0-65535	-
19	Chained test	2 Bytes	0-65535	0 : OFF , 1 : ON
21	Kind of Test *	2 Bytes	0-65535	Table-Kind of test
23	Result of the test *	2 Bytes	0-65535	Table-Results
25	Phase of the test *	2 Bytes	0-65535	Table-Phase of test
27	Time remaining end of test (HIGH) *	2 Bytes	0-65535	-
29	Time remaining end of test (LOW) *	2 Bytes	0-65535	-
31	Unit measure of time *	2 Bytes	0-65535	Table-Measure
33	Decimal points of time *	2 Bytes	0-65535	-
35	Sign of the pressure *	2 Bytes	0-65535	0 : "+", 1 : "-"
37	Pressure at end of test (HIGH) *	2 Bytes	0-65535	-
39	Pressure at end of test (LOW) *	2 Bytes	0-65535	-
41	Unit measure of pressure at end of test *	2 Bytes	0-65535	Table-Measure
43	Decimal points pressure at end of test *	2 Bytes	0-65535	-
45	Sign of decay *	2 Bytes	0-65535	0 : "+", 1 : "-"
47	Decay at end of test (HIGH) *	2 Bytes	0-65535	-
49	Decay at end of test (LOW) *	2 Bytes	0-65535	-
51	Unit measure of the decay at the end of test *	2 Bytes	0-65535	Table-Measure
53	Decimal points decay at end of test *	2 Bytes	0-65535	-
55	Sign of AUX1 decay	2 Bytes	0-65535	0 : "+", 1 : "-"
57	AUX1 Decay at end of test (HIGH)	2 Bytes	0-65535	-
59	AUX1 Decay at end of test (LOW)	2 Bytes	0-65535	-
61	Unit measure of the AUX1 decay end of test	2 Bytes	0-65535	Table-Measure
63	Decimal points AUX1 decay at end of test	2 Bytes	0-65535	-
65	Sign of AUX2 decay	2 Bytes	0-65535	0 : "+", 1 : "-"
67	AUX2 Decay at end of test (HIGH)	2 Bytes	0-65535	-
69	AUX2 Decay at end of test (LOW)	2 Bytes	0-65535	-
71	Unit measure of the AUX2 decay end of test	2 Bytes	0-65535	Table-Measure
73	Decimal points AUX2 decay at end of test	2 Bytes	0-65535	-
75	Sign of the temperature	2 Bytes	0-65535	0 : "+", 1 : "-"
77	Temperature	2 Bytes	0-65535	-
79	Unit of measure of the temperature	2 Bytes	0-65535	Table-Measure
81	Decimal points of the temperature	2 Bytes	0-65535	-

* Parameters that change between channel A and channel B (**This feature is only available on Bi-channel instruments**)

This two result commands can read a **maximum** of 41 registers and a **minimum** of 1 register

Example of reading the **0x41** Holding Register

Tx: 01 03 00 41 00 0A 95 D9

→ Read the last available result

Rx: 01 03 14 00 00 00 01 00 0F 00 33 00 11 00 1B 00 09 00 17 00 01 00 00 AB CF

Example of reading the **0x42** Holding Register

Tx: 01 03 00 42 00 0A 65 D9

Rx: 01 03 14 00 00 00 00 00 0F 00 33 00 11 00 1B 00 09 00 17 00 01 00 00 FA 5F

Explanation of how Results Stack work

Tx: 01 03 00 42 00 29 24 00

Rx: 01 03 52 00 01 00 01 00 0A 00 16 00 16 00 1C 00 09 00 17 00 01 00 00
00 00 00 02 00 1A 00 00 00 00 00 00 3C 00 02 00 00 00 04 24 00 00 00 01
00 FF 01 04 29 6E 00 00 00 03 00 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 53 00 00 7B 80

Tx: 01 03 00 42 00 29 24 00

Rx: 01 03 52 00 01 00 00 00 0A 00 16 00 0A 00 1C 00 09 00 17 00 01 00
00 00 00 00 02 00 1A 00 00 00 00 00 3C 00 02 00 00 00 04 25 00 00 00
01 00 FF 01 46 23 B5 00 00 00 03 00 00 00 00 00 00 00 00 00 00 00 00
00 00 00 00 00 00 00 00 00 00 00 00 53 00 00 A3 6A

The Results Stack work as a LIFO(last in first out) stack so this means that the last test the machine has done is the first that it will be read with command 0x42.

The yellow 16 bit value in the response is the number of results that are not yet transmitted, it is progressive and it will be increased when a new test is done and decreased when a result is read on the machine, so in the example above the one with 1 is the last and the other is the one before.

The green one instead is the lost results, this value will only be increased when the number of outcomes that are not yet transmitted exceed the maximum size of the stack, that is 20, and it will only be reset to 0 when the machine is turned off.

Table - Version Value Mapping

Register Position Hex	Definition	Size Bytes	Reference
1	Serial Number (HI)	2 byte	-
3	Serial Number (LO)	2 byte	-
5	Checksum firmware	2 byte	-
7	Checksum boot	2 byte	-
9	Txxx device	2 byte	
11	Full Scale 1 Press	2 Byte	FS1
13	Full Scale 2 Press	2 Byte	
15	Full Scale 3 Press	2 byte	FS2
17	Full Scale 1 VOUT	2 byte	
19	Full Scale 2 VOUT	2 byte	
21	Full Scale 3 VOUT	2 byte	
23	Power supply voltage	2 byte	
25	Fittings	2 bytes	
27	Type of gas	2 bytes	
29	Pneumatic configuration	2 byte	
31	Device optional configuration	2 byte	
33	Model optional configuration	2 byte	
35	Unit measure calibration pressure	2 byte	
37	Decimal points of calibration pressure	2 byte	
39	Unit measure VOUT	2 byte	
41	Decimal points VOUT	2 byte	
43	Unit measure volume	2 byte	
45	Decimal points volume	2 byte	
47	Unit measure time	2 byte	
49	Decimal points time	2 byte	
51	Unit measure pressure calibration set	2 byte	
53	Decimal points pressure calibration set	2 byte	
55	Unit measure VOUT set	2 byte	
57	Decimal points VOUT set	2 byte	
59	Unit measure volume set	2 byte	
61	Decimal points volume set	2 byte	
63	Unit measure time set	2 byte	
65	Decimal points time set	2 byte	
67	Difference between pressure set and calibration	2 byte	
69	Difference between pressure evaluated and calib	2 byte	
71	Address of the first parameter active on TST menu	2 byte	
73	Address of the first parameter active on SET menu	2 byte	
75	Address of the first parameter active on the CTP menu	2 byte	
77	Address of the first parameter active on the VERSION menu	2 byte	
79	Address of the first parameter active on the CALIBRATION menu	2 byte	
81	Position of the sub menu of the first active param	2 byte	
83	Micro controller ID (HI)	2 byte	
85	Micro controller ID (LO)	2 byte	

The version command can read a **maximum** of 43 registers (86 byte) and a **minimum** of 1 **register**

Table - Barcode Value Mapping

Register Position	Definition	Size Bytes	Reference
0	-	-	-
1	Quantity of characters read from barcode (1..40)	1 Byte	-
2	First character read from barcode	1 Byte	-
3	Second character read from barcode	1 Byte	-
4	Third character read from barcode	1 Byte	-
...	...	...	...
41	40 th character read from barcode	1 Byte	

The barcode can read a **maximum** of 20 registers (40 byte) and a **minimum** of 1 byte

Table - CTP Value Mapping

Register Position	Definition	Size Bytes	Reference
0	-	-	-
1	Number of bad results (HIGH)	1 Byte	-
2	Number of bad results (LOW)	1 Byte	-
3	Number of good results (HIGH)	1 Byte	-
4	Number of good results (LOW)	1 Byte	-
5	Year of the last reset of CTP (HIGH)	1 Byte	-
6	Year of the last reset of CTP (LOW)	1 Byte	-
7	Month of the last reset of CTP	1 Byte	-
8	Day of the last reset of CTP	1 Byte	-
9	Hour of the last reset of CTP	1 Byte	-
10	Minute of the last reset of CTP	1 Byte	-

Table - Status Error

Bit Number	Definition
0	If equal to 1 → FLASH ERROR
1	If equal to 1 → EEPRO ERROR
2	If equal to 1 → FULL SCALE PRESSURE ERROR
3	If equal to 1 → FULL SCALE VOUT ERROR
4	If equal to 1 → OUTPUT ERROR

Table - COM Mode

Position	Definition
0	OFF
1	PC
2	TTY
3	Trace at end of test
4	Barcode
5	Lola printer small
6	Lola printer big

Table - COM Baudrate

Position	Definition
0	2400 bps
1	4800 bps
2	9600 bps
3	19200 bps
4	38400 bps
5	57600 bps
6	115200 bps

Table - Kind of Start

Position	Definition
0	Bimanual
1	Panel excluded
2	From Panel
3	4START
4	BCD+Strobe
5	BCD+200ms

Table - Kind of Test

Value Hex	Definition
0	Leak Test
1	Obstruction Test
2	Flow Test

Table - Result of Test

Value Hex	Definition
0	No result
1	Good
2	Bad
3	Good with reserve
4	Inverse decay
5	Bad reference
6	Bad bell
7	Bad flow under threshold
8	Bad pressure over scale
9	Bad VOUT over scale
10	Bad pressure under tolerance
11	Bad pressure over tolerance
12	Bad pressure not reached the level
13	Abort
14	Bad flow over threshold
15	Bad min filling time
16	Bad result volume reference
17	Not used
18	Not used
19	Bad (differential remote T6961)
98	Abort for automations interrupted
99	Test in progress

Table - Phase of test

Value Dec	Definition
1	Start
2	Delay on start
3	Pre-Filling
4	Pre-Filling
5	Pre-Filling
6	Pre-Filling
7	Discharge pre-filling
8	Discharge pre-filling
9	Discharge pre-filling
10	Discharge pre-filling
11	Filling
12	Filling
13	Reload attempt filling
14	Settling
15	Settling
16	Settling
17	Settling
18	Settling
26	Test phase
30	Obstruction test
43	Volume test
50	End of test with GOOD

Table - Unit measures

Value Hex	Definition
0	Mbar
1	Bar
2	hPa
3	Pa
4	Psi
20	Mbar/s
21	Bar/s
22	hPa/s
23	Pa/s
24	Psi/s
40	Cc/h
41	Cc/min
42	l/h
43	l/min
60	s
61	Min
70	Cc
71	l
80	--
81	%
82	Bps
83	C°
84	Conv/s
85	Prg
86	Chin
87	Chout
88	Volt

Table - OUTAN

Value Hex	Definition
0	Nothing
1	Proportional to pressure (or VOUT)

Table - Language

Value Hex	Definition
0	Italian
1	English
2	Espaniol
3	Francais
4	Deutsch
5	Portuguese

Table - Status

Value	Definition
0	Waiting
1	Test in progress
2	Autozero
3	Discharge
4	Calibration

Table – Sub status

Status Value	Sub status Value	Definition
0	Any	No active sub state
1	Any	Follow the table “Phase of test”
2	0	Not used
2	1	Autozero pressure in progress
2	2	Autozero VOUT in progress
2	3	Autozero pressure error
2	4	Autozero VOUT error
3	0	Not used
3	1	Discharge in progress
3	2	Discharge in progress
3	3	Discharge in progress
3	4	Discharge in progress
4	0	Not used
4	1	Calibration in progress
4	2	Calibration in progress
5	0	Not used
5	1	Not used

Table – Connection type

Value	Definition
0	Off
1	Bluetooth 2.0
2	Bluetooth low energy
3	Wi-Fi
4	Hotspot
5	Wi-Fi ForTest

TST Programs Mapping

Program N.	Real Address Hex	Alias Address Hex
0	3322	9D00
1	3362	9D20
2	33A2	9D40
3	33E2	9D60
4	3422	9D80
5	3462	9DA0
6	34A2	9DC0
7	34E2	9DE0
8	3522	9E00
9	3562	9E20
10	35A2	9E40
11	35E2	9E60
12	3622	9E80
13	3662	9EA0
14	36A2	9EC0
15	36E2	9EE0
16	3722	9F00
17	3762	9F20
18	37A2	9F40
19	37E2	9F60
20	3822	9F80
21	3862	9FA0
22	38A2	9FC0
23	38E2	9FE0
24	3922	A000
25	3962	A020
26	39A2	A040
27	39E2	A060
28	3A22	A080
29	3A62	A0A0
30	3AA2	A0C0
31	3AE2	A0E0
32	3B22	A100
33	3B62	A120
34	3BA2	A140
35	3BE2	A160
36	3C22	A180
37	3C62	A1A0
38	3CA2	A1C0

39	3CE2	A1E0
40	3D22	A200
41	3D62	A220
42	3DA2	A240
43	3DE2	A260
44	3E22	A280
45	3E62	A2A0
46	3EA2	A2C0
47	3EE2	A2E0
48	3F22	A300
49	3F62	A320
50	413D	A340
51	417D	A360
52	41BD	A380
53	41FD	A3A0
54	423D	A3C0
55	427D	A3E0
56	42BD	A400
57	42FD	A420
58	433D	A440
59	437D	A460
60	43BD	A480
61	43FD	A4A0
62	443D	A4C0
63	447D	A4E0
64	44BD	A500
65	44FD	A520
66	453D	A540
67	457D	A560
68	45BD	A580
69	45FD	A5A0
70	463D	A5C0
71	467D	A5E0
72	46BD	A600
73	46FD	A620
74	473D	A640
75	477D	A660
76	47BD	A680
77	47FD	A6A0
78	483D	A6C0
79	487D	A6E0
80	48BD	A700
81	48FD	A720
82	493D	A740
83	497D	A760

84	49BD	A780
85	49FD	A7A0
86	4A3D	A7C0
87	4A7D	A7E0
88	4ABD	A800
89	4AFD	A820
90	4B3D	A840
91	4B7D	A860
92	4BBD	A880
93	4BFD	A8A0
94	4C3D	A8C0
95	4C7D	A8E0
96	4CBD	A900
97	4CFD	A920
98	4D3D	A940
99	4D7D	A960
100	4DBD	A980
101	4DFD	A9A0
102	4E3D	A9C0
103	4E7D	A9E0
104	4EBD	AA00
105	4EFD	AA20
106	4F3D	AA40
107	4F7D	AA60
108	4FBD	AA80
109	4FFD	AAA0
110	503D	AAC0
111	507D	AAE0
112	50BD	AB00
113	50FD	AB20
114	513D	AB40
115	517D	AB60
116	51BD	AB80
117	51FD	ABA0
118	523D	ABC0
119	527D	ABE0
120	52BD	AC00
121	52FD	AC20
122	533D	AC40
123	537D	AC60
124	53BD	AC80
125	53FD	ACA0
126	543D	ACC0
127	547D	ACE0
128	54BD	AD00

129	54FD	AD20
130	553D	AD40
131	557D	AD60
132	55BD	AD80
133	55FD	ADA0
134	563D	ADC0
135	567D	ADE0
136	56BD	AE00
137	56FD	AE20
138	573D	AE40
139	577D	AE60
140	57BD	AE80
141	57FD	AEA0
142	583D	AEC0
143	587D	AEE0
144	58BD	AF00
145	58FD	AF20
146	593D	AF40
147	597D	AF60
148	59BD	AF80
149	59FD	AFA0
150	5A3D	AFC0
151	5A7D	AFE0
152	5ABD	B000
153	5AFD	B020
154	5B3D	B040
155	5B7D	B060
156	5BBD	B080
157	5BFD	B0A0
158	5C3D	B0C0
159	5C7D	B0E0
160	5CBD	B100
161	5CFD	B120
162	5D3D	B140
163	5D7D	B160
164	5DBD	B180
165	5DFD	B1A0
166	5E3D	B1C0
167	5E7D	B1E0
168	5EBD	B200
169	5EFD	B220
170	5F3D	B240
171	5F7D	B260
172	5FBD	B280
173	5FFD	B2A0

174	603D	B2C0
175	607D	B2E0
176	60BD	B300
177	60FD	B320
178	613D	B340
179	617D	B360
180	61BD	B380
181	61FD	B3A0
182	623D	B3C0
183	627D	B3E0
184	62BD	B400
185	62FD	B420
186	633D	B440
187	637D	B460
188	63BD	B480
189	63FD	B4A0
190	643D	B4C0
191	647D	B4E0
192	64BD	B500
193	64FD	B520
194	653D	B540
195	657D	B560
196	65BD	B580
197	65FD	B5A0
198	663D	B5C0
199	667D	B5E0
200	66BD	B600
201	66FD	B620
202	673D	B640
203	677D	B660
204	67BD	B680
205	67FD	B6A0
206	683D	B6C0
207	687D	B6E0
208	68BD	B700
209	68FD	B720
210	693D	B740
211	697D	B760
212	69BD	B780
213	69FD	B7A0
214	6A3D	B7C0
215	6A7D	B7E0
216	6ABD	B800
217	6AFD	B820
218	6B3D	B840

219	6B7D	B860
220	6BBD	B880
221	6BFD	B8A0
222	6C3D	B8C0
223	6C7D	B8E0
224	6CBD	B900
225	6CFD	B920
226	6D3D	B940
227	6D7D	B960
228	6DBD	B980
229	6DFD	B9A0
230	6E3D	B9C0
231	6E7D	B9E0
232	6EBD	BA00
233	6EFD	BA20
234	6F3D	BA40
235	6F7D	BA60
236	6FBD	BA80
237	6FFD	BAA0
238	703D	BAC0
239	707D	BAE0
240	70BD	BB00
241	70FD	BB20
242	713D	BB40
243	717D	BB60
244	71BD	BB80
245	71FD	BBA0
246	723D	BBC0
247	727D	BBE0
248	72BD	BC00
249	72FD	BC20
250	733D	BC40
251	737D	BC60
252	73BD	BC80
253	73FD	BCA0
254	743D	BCC0
255	747D	BCE0
256	74BD	BD00
257	74FD	BD20
258	753D	BD40
259	757D	BD60
260	75BD	BD80
261	75FD	BDA0
262	763D	BDC0
263	767D	BDE0

264	76BD	BE00
265	76FD	BE20
266	773D	BE40
267	777D	BE60
268	77BD	BE80
269	77FD	BEA0
270	783D	BEC0
271	787D	BEE0
272	78BD	BF00
273	78FD	BF20
274	793D	BF40
275	797D	BF60
276	79BD	BF80
277	79FD	BFA0
278	7A3D	BFC0
279	7A7D	BFE0
280	7ABD	C000
281	7AFD	C020
282	7B3D	C040
283	7B7D	C060
284	7BBD	C080
285	7BFD	C0A0
286	7C3D	C0C0
287	7C7D	C0E0
288	7CBD	C100
289	7CFD	C120
290	7D3D	C140
291	7D7D	C160
292	7DBD	C180
293	7DFD	C1A0
294	7E3D	C1C0
295	7E7D	C1E0
296	7EBD	C200
297	7EFD	C220
298	7F3D	C240
299	7F7D	C260

Exceptions

Exception code	Definition
0	-
1	Command not supported
2	Illegal data address
3	Illegal data value
4	Illegal function

All responses with a function code > 0x80h are exceptions

Function code 0x81h is an exception response of the request with function code 0x80h

Function code 0x82h is a exception response of the request with function code 0x81h

CRC16

Example of the CRC16 algorithm

```
Public Function CRC16(ByVal Buffer() As Byte, ByVal length As Integer) As Long
    Dim I As Long
    Dim CRC As Long = &HFFFF
    Dim LSB As Long = 0
    Dim J As Integer
    For I = 0 To length
        CRC = CRC Xor Buffer(I)
        For J = 0 To 7
            LSB = CRC And &H1
            CRC = Math.Floor(CRC / 2)
            If (LSB) Then
                CRC = (CRC Xor &HA001)
            End If
        Next J
    Next I
    Return CRC
End Function
```

Modpoll Modbus Master Simulator

Modpoll is a free command line based Modbus master simulator and test utility.

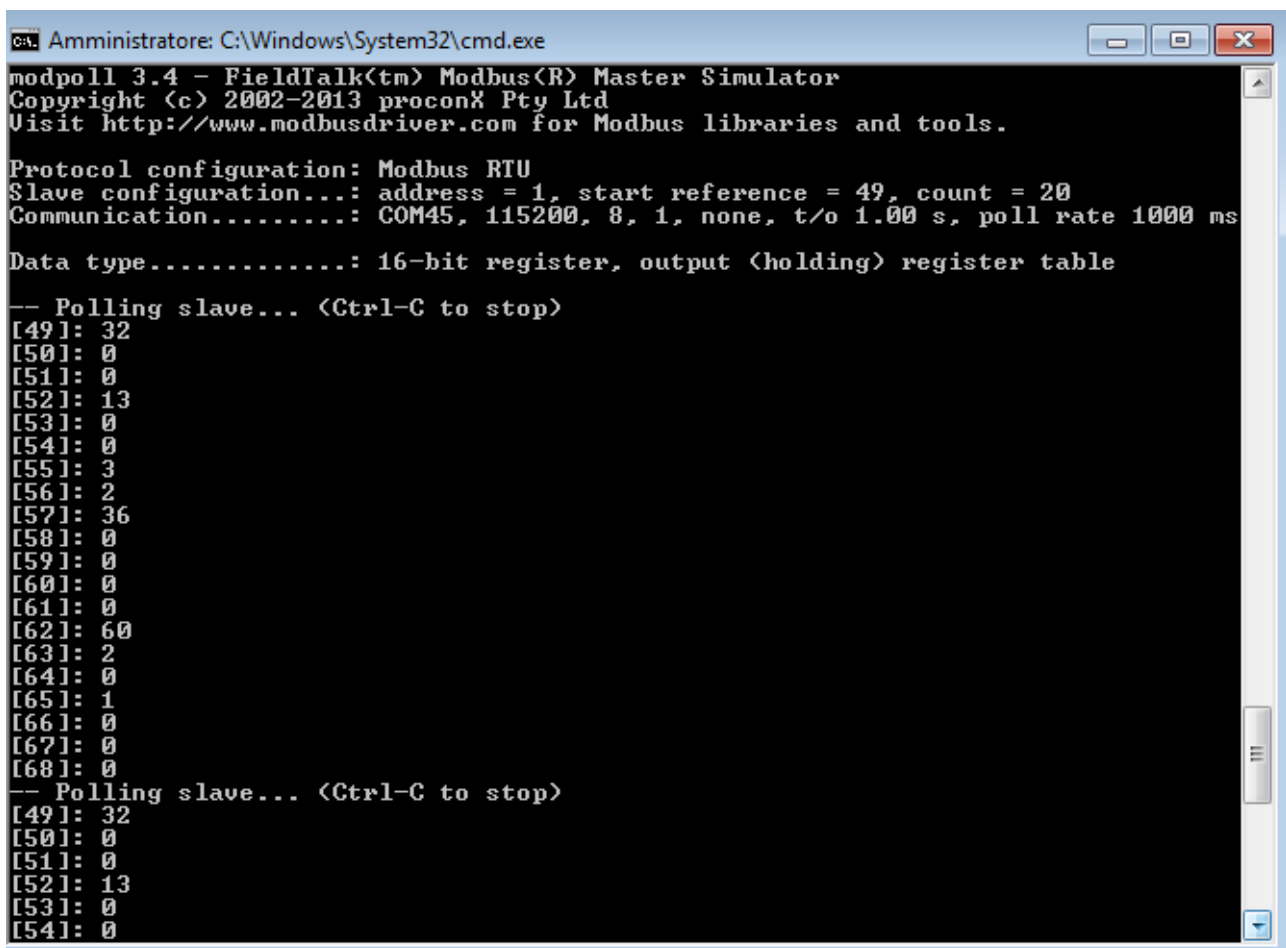
Download available here <http://downloads.fortest-leak-testing.it/modpoll.zip>

Binaries are available for the following operating system platforms and architectures:

- Windows
- Linux (x86 32-bit)
- QNX Neutrino 6 (x86)

This example implement a polling request every 1000ms via MODBUS of the command status (0x30 -> 49 decimal, please note that address here are shifted 1+) using the READ HOLDING REGISTER COMMAND.

Example: `modpoll -b 115200 -d 8 -s 1 -p none -a 1 -r 49 -c 20 -l 1000 COM45`



```
Amministratore: C:\Windows\System32\cmd.exe
modpoll 3.4 - FieldTalk(tm) Modbus(R) Master Simulator
Copyright (c) 2002-2013 proconX Pty Ltd
Visit http://www.modbusdriver.com for Modbus libraries and tools.

Protocol configuration: Modbus RTU
Slave configuration...: address = 1, start reference = 49, count = 20
Communication.....: COM45, 115200, 8, 1, none, t/o 1.00 s, poll rate 1000 ms
Data type.....: 16-bit register, output (holding) register table

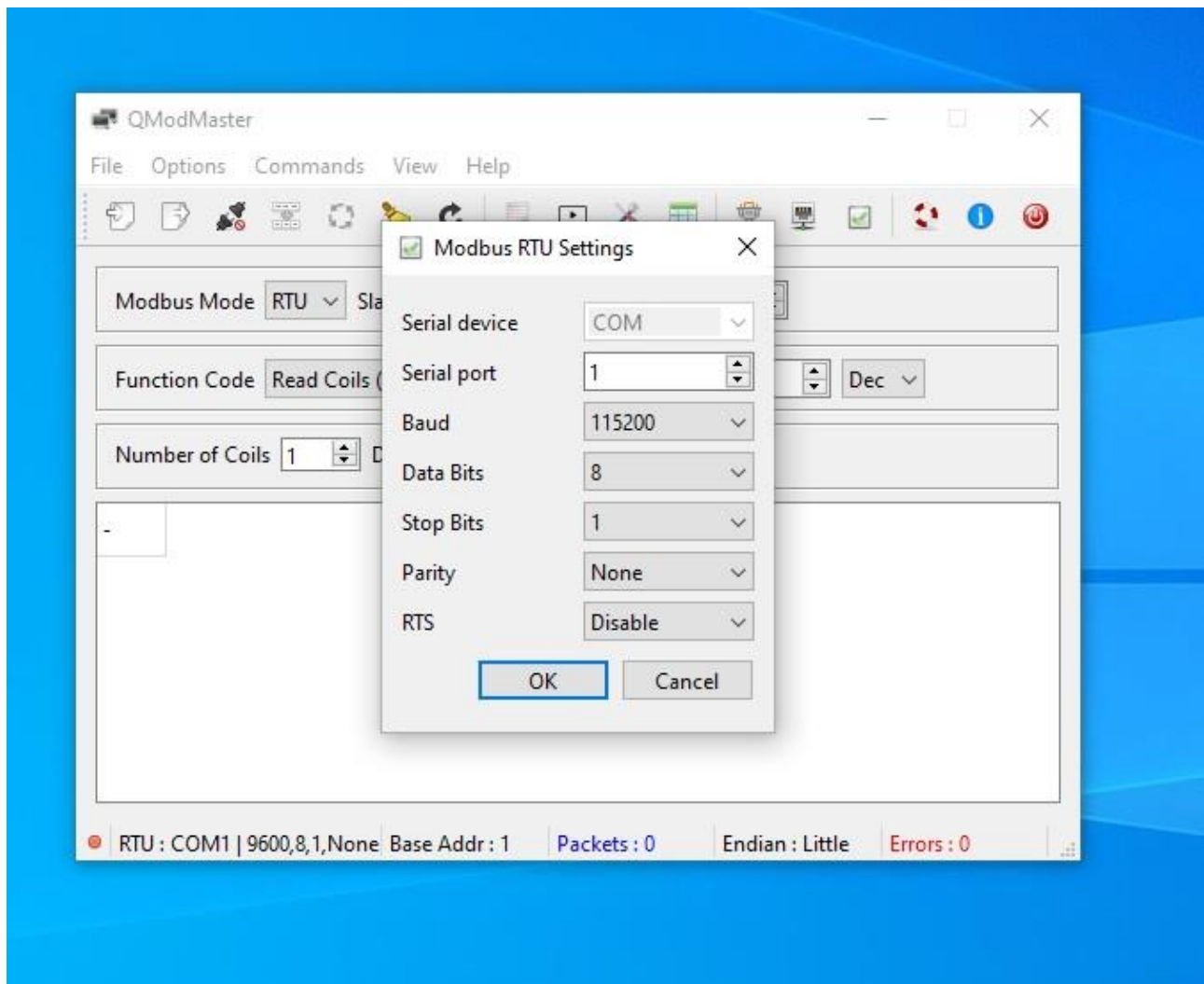
-- Polling slave... (Ctrl-C to stop)
[49]: 32
[50]: 0
[51]: 0
[52]: 13
[53]: 0
[54]: 0
[55]: 3
[56]: 2
[57]: 36
[58]: 0
[59]: 0
[60]: 0
[61]: 0
[62]: 60
[63]: 2
[64]: 0
[65]: 1
[66]: 0
[67]: 0
[68]: 0
-- Polling slave... (Ctrl-C to stop)
[49]: 32
[50]: 0
[51]: 0
[52]: 13
[53]: 0
[54]: 0
```

Please note that with values greater than 65535 (for example pressure, decay and time elapsed) modpoll will show wrong values

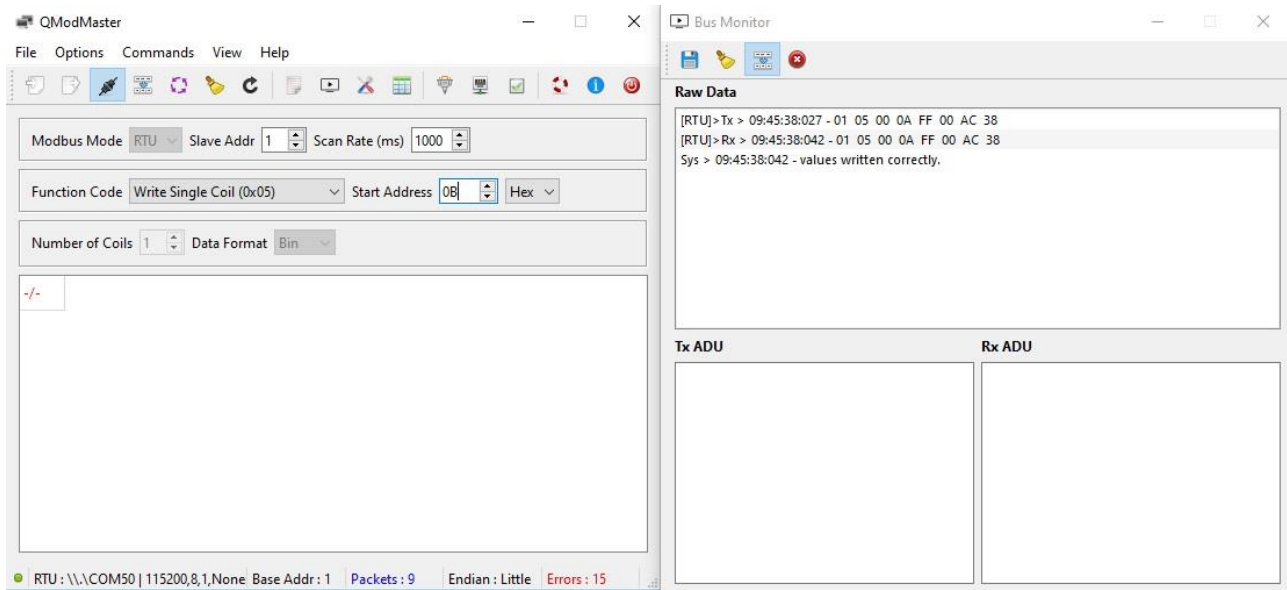
qModbus Master Simulator

qModMaster is a windows application that allow to execute Modbus RTU commands

Download available here <https://sourceforge.net/projects/qmodmaster/files/>



Settings menu (COM port depends from available COM ports)



Example of START signal

Example application for PLC Siemens

- **Example application source code for Siemens S7**
<http://downloads.fortest-leak-testing.it/MODBUS/Progetto Eempio PLC Test Modbus V13.zip>
- **General guidelines for interfacing PLC with ForTest (Italian)**
http://downloads.fortest-leak-testing.it/MODBUS/Comunicazione Modbus PLC S7 1200_rev02.pdf

Modbus RTU over Ethernet

Modbus protocol over Ethernet is supported only in "Encapsulated RTU over TCP" mode.

Transactions are not supported

```
d:\MODBUS>modpoll -p 502 -o 1 -l 500 -m enc -a 1 -c 20 -r 49 192.168.0.41
modpoll 3.4 - FieldTalk(tm) Modbus(R) Master Simulator
Copyright (c) 2002-2013 proconX Pty Ltd
Visit http://www.modbusdriver.com for Modbus libraries and tools.

Protocol configuration: Encapsulated RTU over TCP
Slave configuration...: address = 1, start reference = 49, count = 20
Communication.....: 192.168.0.41, port 502, t/o 1.00 s, poll rate 500 ms
Data type.....: 16-bit register, output (holding) register table

-- Polling slave... (Ctrl-C to stop)
[49]: 32
[50]: 0
[51]: 0
[52]: 13
[53]: 0
[54]: 1
[55]: 3
[56]: 2
[57]: 35
[58]: 0
[59]: 0
[60]: 0
[61]: 0
[62]: 60
[63]: 2
[64]: 0
[65]: 0
[66]: 0
[67]: 0
[68]: 2
-- Polling slave... (Ctrl-C to stop)
[49]: 32
[50]: 0
[51]: 0
[52]: 13
[53]: 0
[54]: 1
[55]: 3
[56]: 2
[57]: 35
[58]: 0
[59]: 0
[60]: 0
[61]: 0
[62]: 60
[63]: 2
[64]: 0
[65]: 0
[66]: 0
[67]: 0
[68]: 2
-- Polling slave... (Ctrl-C to stop)
```

Changelog

DATE	DESCRIPTION	REVISION
19/06/2014	First revision of this document	Rev01
18/07/2014	Correct the write single register example , insert the changelog , correct the TST program mapping, table version checksum size .	Rev02
21/07/2014	Address correction in the SET menu table, insert new tables (OUTAN, Language)	Rev03
22/07/2014	Update the size on the table "Result Value Mapping" and insert an example	Rev03
05/09/2014	Add the Status and Sub status reference table Insert a reference in the Status value mapping to the Result table	Rev04
29/06/2016	Added some informations in introduction and in some examples	Rev05
06/03/2018	Added modpoll modbus master simulator example Examples update and corrections Corrections in CRC and command definition	Rev06
03/07/2018	Added example application and interfacing guidelines for PLC	Rev07
12/09/2018	Changed link of example application	Rev 08
05/10/2018	Modified title of the document	Rev09
24/10/2018	Added Modbus RTU over Ethernet , reduce EEPROM write attempts , examples	Rev10
25/01/2019	Updated the results table	Rev11
08/07/2019	Update on status table	Rev12
16/06/2022	Added new supported commands on latest firmware release (r/w multiple) Added new example on results commands transitions	Rev13
17/06/2022	Added new examples Added bi-channel T8890 additional commands	Rev14
19/09/2023	Update appendix table and alias addresses	Rev15
29/10/2023	Update appendix table unit measure	Rev16
02/10/2023	Added stack command and examples	Rev17

For any technical inquiries or error reports, please contact support@fortest.com